

TABLE OF CONTENTS

S-250 TEMPORARY ROAD WORKS

	Page No.
1. GENERAL	S.250-2
2. SUBGRADE PREPARATION.....	S.250-2
3. SUBBASE AND BASE COURSES	S.250-3
3.1 General.....	S.250-3
3.2 Material	S.250-3
3.3 Construction of Subbase Course (30cm thick).....	S.250-4
3.4 Construction Base Course (20 cm thick)	S.250-4
3.5 Asphalt Primer/ Primer Coat.....	S.250-5
4. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC).....	S.250-5

S.250 TEMPORARY ROAD WORKS

1. GENERAL

The scope of work under this section covers the furnishing of all materials, labor, equipment, and transportation and installation service to prepare the sub-grade for roads placing the crushed stones sub-base, base and surface courses and asphalt surfacing as shown on the approval drawings.

The work shall include some protection and support required to existing structure, such as: ditch, duct and others underground structures.

The existing roads in the site boundary to be used during the construction is damaged, the Bidder shall repair them. The cost for repairing existing road shall be included in the Contract Price.

2. SUBGRADE PREPARATION

- 2.1 Excavate subgrade or fill, as required, and then graded and compacted to provide a firm foundation of uniform density throughout. Completed subgrade shall be true to alignment, grade and cross section, including the required crown as shown on the drawings. CBR test on the furnished level of completed subgrade shall have a minimum value of 6 in the condition 98% dry density.
- 2.2 Prior to placing of any fill required to bring the subgrade to its finished level, strip areas to be covered of any loose or otherwise unapproved fill materials, organic materials, or any foreign or deleterious matter.
- 2.3 Holes, ruts and similar defects shall be filled with approved materials. Unstable areas, projecting stone or rock, and similar defects shall be cut out and the holed areas shall be filled.
- 2.4 All fill required shall be compacted as specified in the Technical Specification-Earth Works (S300)

- 2.5 Sub base preparation for roads should be started only when the required of CBR subgrade has been achieved.
- 2.6 Drainage culvert under the subgrade shall have been constructed before the subgrade works are carried out.
- 2.7 To enable the Bidder to carry out the work during wet season, temporary drainage shall be provided to drain smoothly.

3. SUBBASE AND BASE COURSES

3.1 General

The Bidder shall prepare the sub-grade and apply the sub-base and base course materials for roads as specified herein.

3.2 Material

Material used for the sub-base and base course shall consist of materials as specified below.

3.2.1 Sub-base Material

Material shall be obtained from a source approved by the Owner.

This material shall consist of well graded crushed stone with hard. Strong, durable rock pieces having the following gradation (size C-40 ,JIS A 5001), and equivalent of it.

Name of crushed stone: Crushed run, C-40

<u>Sieve Size</u> (mm)	<u>Percentage Passing by Weight</u> (%)
53	100
37.5	95~100
19	50~80
4.75	15~40
2.36	5~25

3.2.2 Base Course Material

This material shall consist of well graded crushed stone with hard, strong, durable pieces having the following gradation (size C-40, JIS A 5001), and equivalent of it.

<u>Sieve Size</u> (mm)	<u>Percentage Passing by Weight</u> (%)
53	100
37.5	95~100
19	60~90
4.75	30~65
2.36	20~50
425µ m	10~30
75µ m	2~10

3.3 Construction of Subbase Course (30cm thick)

3.3.1 Immediately before placing the subbase material, the surface shall be proof rolled and compacted to a dry density not less than 98 percent of the maximum dry density as determined by the method specified in ASTM D698, Method D.

3.3.2 Any soft, unsound areas, encountered during the preparation and compaction of the surfaces, shall be removed. The unsound areas, pockets or cavities, shall be excavated and backfilled with approved material to the depth and extend beyond the neat lines and elevations as shown on the drawings or as may be directed by the Owner.

3.3.3 Shaping and compaction shall be carried out concurrently so that upon completion, the surfaces shall be smooth and everywhere within plus or minus 30 mm of the required grade, but not uniformly high or low.

3.3.4 The surface shall be re-graded and re-compacted at the Bidder's expenses, if disturbed by any reason before or during the placing of the base course material.

3.4 Construction Base Course (20 cm thick)

- 3.4.1 After the subbase for the road has been completed, the Bidder shall place the base course material in layers not less than 100mm of compacted thickness.
- 3.4.2 Road bases and surfaces shall be compacted to a dry density not less than 98 percent of the maximum dry density as determined by the method specified in ASTM D698, Method D. Water shall be added during compaction as required to achieve the specified dry density.
- 3.4.3 Road surfaces shall be shaped and compacted to the lines, grades and dimensions shown on the drawings by equipment designed for that purpose. Such shaping and compaction shall be carried out concurrently so that upon completion the surfaces shall be smooth and everywhere within plus and minus 15mm of required grade, and shall be compacted to the degree specified for the full depth. Any local depressions which do not meet the above tolerances shall be filled with approved granular surface materials.
- 3.5 **Asphalt Primer/ Primer Coat**
- 3.5.1 The crushed rock base course shall be dried, swept clean of all dust, dirt and loose particles and prepared for surfacing.
A prime coat of asphalt primer ASTM D2027 Grade MC70 shall be sprayed over to the base course at a rate 0.90 liter per square meter required to fill all voids and interstices and uniformly coat the base course surface.
- 3.5.2 Asphalt paving will be constructed on the section of Finishing.
4. **Quality Assurance/Quality Control (QA/QC)**
- 4.1 Quality assurance/quality control shall be in accordance with the requirements of Part 4, Sub-section 2.7 and this Technical Specification.
- 4.2. Quality Control procedures and Documentation

(Last page of S-250)